Willow WU

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Software Engineer — Backend / Full-Stack + AI Systems — Python/TypeScript · Distributed Systems · LLM & GNN

EDUCATION

Columbia University *New York, NY*
M.S. in Computer Science *(Machine Learning Track)* — GPA 3.96/4.0 *Sep 2025 – Dec 2026*
Relevant Coursework: Analysis of Algorithms, Applied Machine Learning, Deep Learning, Natural Language Processing, Engineering Software-as-a-Service (SaaS), Advanced Spoken Language Processing

Hong Kong Baptist University (HKBU) *Hong Kong*
B.Sc. in Data Science *Sep 2021 – Jun 2025*

EXPERIENCE

Microsoft — Software Engineer Intern *Summer 2026*
OfficePLUS AI Assistant

- Contributed to refactoring a multi-session, token-streaming LLM chat engine (React/TypeScript, WebSocket RPC), consolidating **8** run-tracking tables into a 2-table session model and eliminating active-run eviction, a state-corruption failure mode implicated in ∼**40%** of user downvotes.
- Built a routing-verification probe proving **500/500** concurrent runs routed correctly with **zero** cross-session cross-talk; audited 6 fallback branches to prevent silent routing regressions.
- Diagnosed production failures across **3 non-joinable telemetry stores** (Azure SQL + 2 App Insights tenants) with KQL and T-SQL over **11.5K+** events, proving **97.6%** of "rate-limit" failures were genuine quota exhaustion and correcting the team's leading hypothesis.

Columbia Business School — Research Assistant *New York, NY*
Prof. Emmanuel Yimfor — VC/PE Founder-Diversity Research *Jun 2026 – Present*

- Built an end-to-end Python/DuckDB ML pipeline to infer race labels for **419,741** no-photo LinkedIn profiles, joining a **962,757-row** roster with **543,016** image-labeled profiles.
- Designed a multiclass classifier over name char/word n-grams and fold-safe target-encoded geographic, education, prestige, and connection features, lifting macro-F1 from **0.71** to **0.79** on 5-fold out-of-fold evaluation.
- Engineered CBS GRID HPC execution (SGE, scratch/shared storage, Parquet intermediates, DuckDB SQL joins), replacing a costly full nested-Parquet merge with a targeted full-coverage feature join.

Columbia University — Teaching Assistant, Introduction to Databases *New York, NY*
Prof. Kenneth Ross *Sep 2025 – Present*

- Mentored across a **300-student** databases course, advising **6 project teams (∼40 students)** on relational schema design, SQL, and query-plan analysis.
- Graded midterms, finals, and assignments and debugged student queries for execution plans, indexing strategies, and scalable data-system design.

Shenzhen Haohai Zhiyuan — Software Engineer Intern *Shenzhen, China*
Nov 2023 – Mar 2024

- Built AWS-backed RESTful services and React Native / Flutter cross-platform interfaces for a hardware-integrated product, shipping multilingual support across 10+ languages.
- Designed automated IoT device-testing and monitoring pipelines with the hardware team, improving telemetry accuracy and release reliability while cutting manual QA.

PROJECTS

Evidence of Life — AI Life-Tracking Web App *Startup Studio* *2026*

- Built a solo full-stack React/TypeScript SPA on **Supabase** (Auth, Postgres, Edge Functions) with AI-assisted input classification, Google Calendar sync, and map-based place memory, backed by **51** passing Vitest tests.

ML Inference Service — FastAPI, Docker, batch inference *Personal* *2026*

- Built a Dockerized FastAPI inference service with single-record and batch scoring, health checks, and a swappable model interface; benchmarked ∼**17 ms** p50 / ∼**30 ms** p95 latency and ∼**78K records/s** batch throughput.

PUBLICATIONS

Explainable Dynamic Graph Neural Networks for Predictive Maintenance in Vehicle Chassis Systems *First Author* *2025*
You Wu, Junwei Su, Sirui Zhang, Zhijian Li. Proceedings of Machine Learning Research (PMLR), **ACML 2025**.

- Developed a graph-based failure-prediction system (**NRP-GCN-RF**: GCN encoder + Random Forest) over weighted fault graphs on ∼9M records: **98.9%** fault-category accuracy and high-cost failure precision **60%** → **95%** under leakage-aware cross-validation.

SKILLS

Languages: Python, TypeScript/JavaScript, SQL, Ruby, C++
Web / Backend: React, Node.js, FastAPI, Ruby on Rails, REST APIs, WebSocket RPC, Supabase, Tailwind, shadcn/ui
Data / ML: PyTorch, scikit-learn, pandas, DuckDB, PostgreSQL, Elasticsearch, Graph Neural Networks (GNNs), Deep Learning, Transformers, NLP, Large Language Models (LLM/RAG), ETL
Infra / Foundations: Azure (KQL, Azure SQL, DevOps), Docker, CI/CD, HPC, Distributed Systems; RSpec/Vitest; Data Structures & Algorithms, OOP

HONORS & AWARDS

Kaggle Silver Medal (Top 4.2%) • **MCM Honorable Mention** (Top 20%) • **CUMCM Provincial 2nd Prize** • Merit Scholarship (Top 15%)